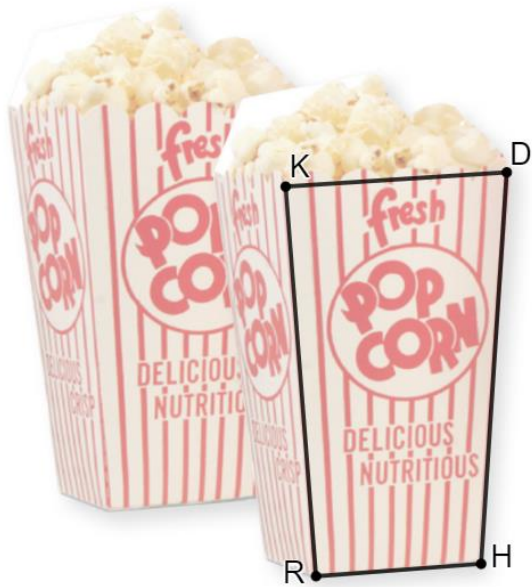
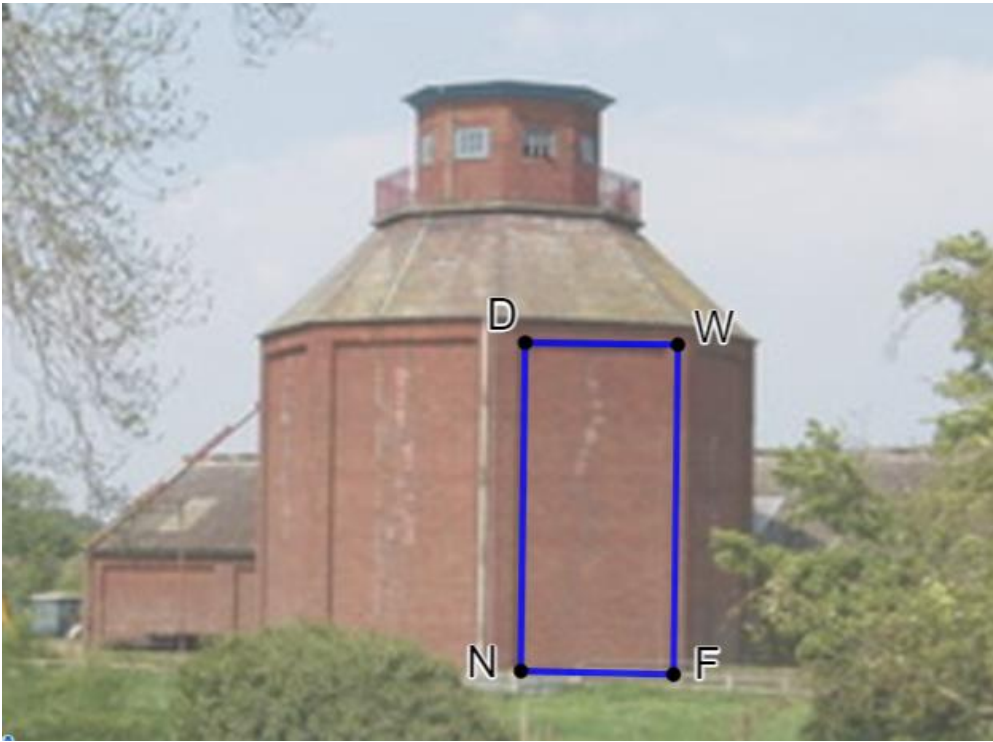


8.6.2 Activity

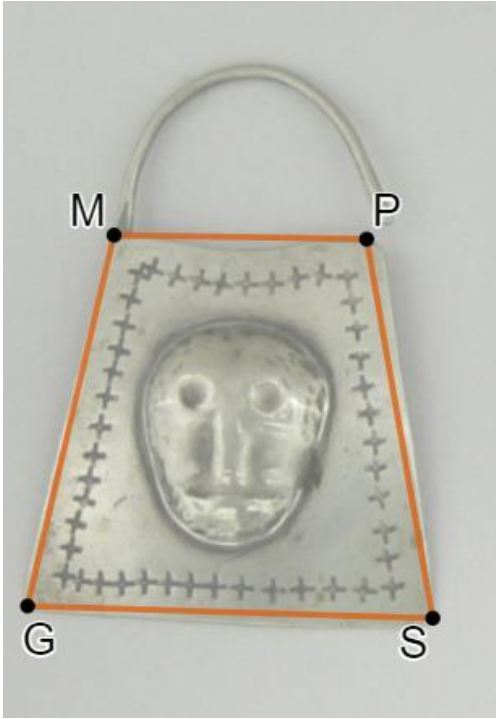
1. The local movie theater is creating a new design on the face of the popcorn box. Isosceles trapezoid $KDHR$ has side lengths $KR = \left(\frac{1}{2}x + 5\right)$ inches (in) and $DH = (2x - 4)$ in. What is the measurement of $\overline{KR}$?



2. The owner of the barn wants to paint each rectangular panel. He found the measurements of rectangular panel $DWFN$. $DN = (12x + 4)$ feet (ft), $WF = (20x - 8)$ ft, $DW = (5y - 5)$ ft, and $NF = (2y + 4)$. What is the area of one rectangular panel?



3. Shauna makes earrings to sell at the farmer's market. She cuts the metal of each earring using the same isosceles trapezoid model $MPSG$, as shown, where $m\angle MGS = (8x - 10)^\circ$, $m\angle PSG = (6x + 12)^\circ$, $m\angle GMP = (22y - 19)^\circ$, and $m\angle MPS = (16y + 14)^\circ$.



If Shauna creates three similar sizes using the model, what is the measure of each angle?

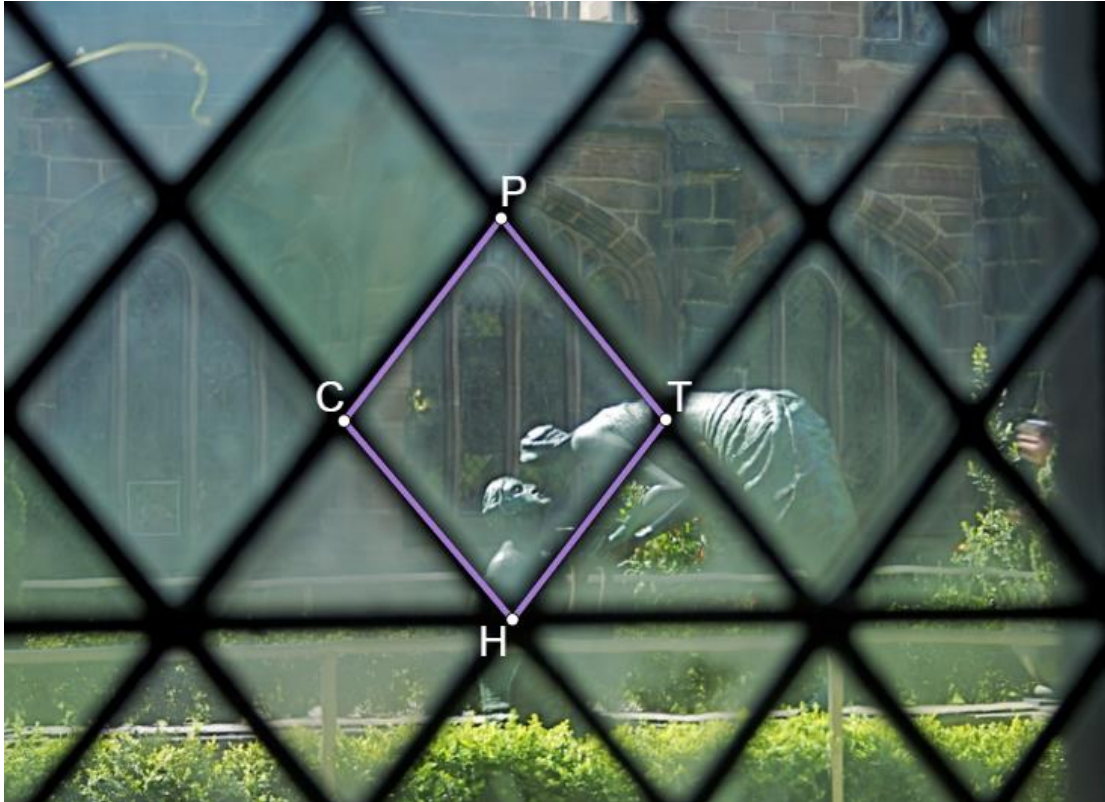
$$m\angle MGS =$$

$$m\angle PSG =$$

$$m\angle GMP =$$

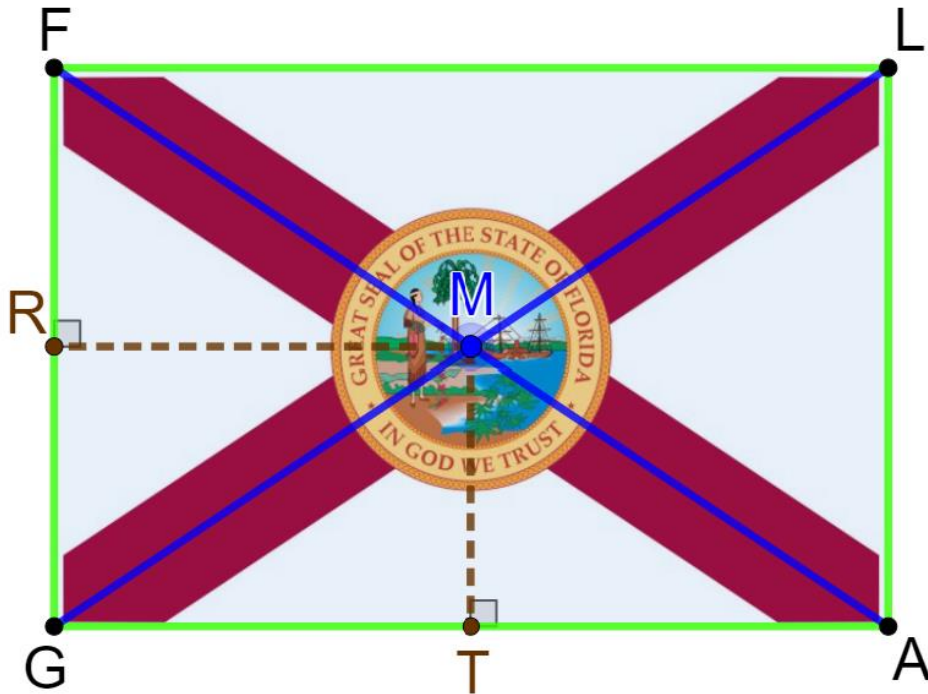
$$m\angle MPS =$$

4. The Chester Cathedral in England has windows looking into the courtyard at the Water for Life statue. The window glass consists of multiple rhombus panels. Rhombus $PTHC$ is shown, where $m\angle PCH = (6x - 23)^\circ$, $m\angle PTH = (4x + 19)^\circ$, and $m\angle CPT = (8y - 23)^\circ$. What is the value of y ?



5. Ha-joon was painting the Florida flag onto a rectangular piece of wood. He needed to find the precise center so that the Seal of Florida was in the center of the piece of wood. He has made a drawing to help determine how far to measure from each vertex and from the midpoint of each side the seal should be placed.

Rectangle $FLAG$ is shown, where $FM = (3x + 20)$ in., $MA = (5x - 3)$ in., $FL = (8x - 1)$ in., and $FG = (5x + 2.5)$ in.



Find the measurement of each. When necessary, round to the nearest thousandth.

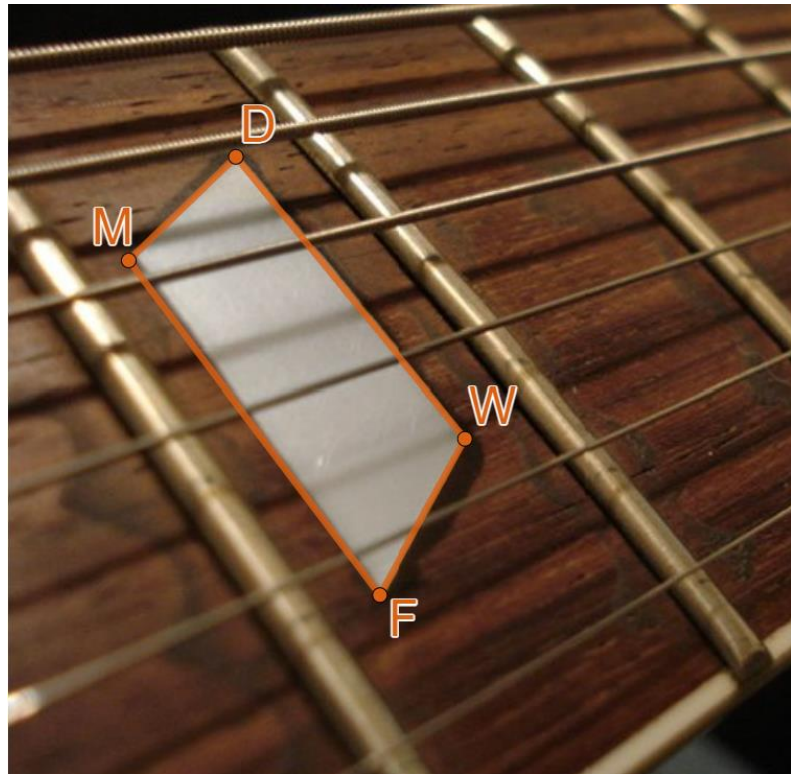
$$FM =$$

$$RM =$$

$$TM =$$

6. Fret markers on a guitar are both decorative and purposeful. The markers are visual clues to the performer. The fret marker shown is labeled *DWFM*.

Determine whether *DWFM* is a parallelogram given the following measurements:



$$m\angle DMF = (13x + 4.4)^\circ, m\angle MDW = (10x + 13)^\circ$$
$$m\angle DWF = (15x + 4)^\circ, m\angle WFM = (10x - 7)^\circ$$

7. A rectangular window is shown with 8 congruent rectangular windowpanes. $KRWF$ represents the entire window and $CHXF$ represents one pane.



If $KR = 4 \cdot CH$, $RW = 2 \cdot HX$, $KR = 11x + 19$ centimeters (cm), $WR = (9x + 3)$ cm, $CH = (3x + 3)$ cm, and $HX = (5x - 2)$ cm, find the dimensions of $KRWF$.

8. The facade of the Washington Monument is an isosceles trapezoid, as shown. Determine the measure of $\overline{DW}$ given the following measurements, in meters (m).

$$XW = (12x - 15) \text{ m}$$

$$DN = (9x + 27) \text{ m}$$

$$XN = (8x + 40.5) \text{ m}$$

